

Formulating Interfacial Impedances for Designing High-Energy and High-Power All-Solid-State Battery Cathodes

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Cite This: *ACS Appl. Mater. Interfaces* 2024, 16, 26066–26078



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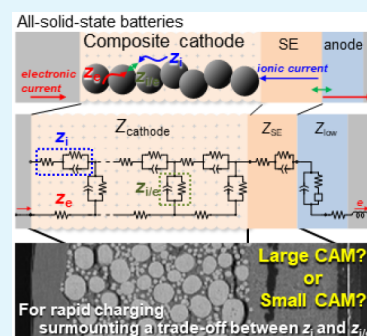
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ABSTRACT: All-solid-state batteries (ASSBs) are safe, high-energy-storage systems. However, despite the progress achieved in the development of high-ionic-conductivity solid electrolytes (SEs), the power performance of ASSBs remains low because of the high interfacial impedances in composite cathodes. Therefore, understanding the interfacial factors is crucial for obtaining high power ASSBs. This study provides a quantitative analysis of the influence of these factors using impedance spectroscopy measurements, which enables the elucidation of the interfacial impedance values of two key parameters, the grain-boundary resistance ($r_{i,gb}$) and charge-transfer resistance ($r_{i/e}$). Systematic investigation revealed an unexpected increase in the cathodic resistance with the decrease in the size of the cathode active material (CAM) particles, indicating that even high-reaction-surface-area CAMs yield low $r_{i/e}$ but high $r_{i,gb}$ values owing to their high porosity, resulting in a trade-off relationship. In contrast, this phenomenon is unlikely to occur in liquid-electrolyte-based batteries. Notably, we discuss how composite cathode design impacts performances of stable, high-power, and high-energy ASSBs.

KEYWORDS: all-solid-state battery, interfacial impedance, composite cathode, electrochemical impedance spectroscopy, grain-boundary resistance, charge-transfer resistance



INTRODUCTION

All-solid-state batteries (ASSBs) are attracting considerable attention for use in high-energy-storage systems ($>1000 \text{ W}\cdot\text{h}\cdot\text{kg}^{-1}$) having an anode-free cell structure.^{1,2} Inorganic solid electrolytes (SEs)—such as oxides,³ halides,⁴ and borohydrides⁵—are key materials for ASSBs, and sulfide-based SEs (SSEs) notably enhance ASSBs as a result of their ductility⁶ and high ionic conductivity^{1,7–9} ($>1 \text{ mS}\cdot\text{cm}^{-1}$) compared to organic liquid electrolytes (LEs). However, despite the high ionic conductivity of SSEs, the room-temperature power performance of ASSBs remains unsatisfactory owing to the high-resistance components resulting from the solid–solid interfacial contact.¹⁰

ASSBs contain a thick composite cathode consisting of a mixture of SSE particles and cathode active material (CAM) particles, in which solid-specific microstructural factors result in highly resistive charge-transport processes.¹⁰ Notably, the solid–solid contact of SSE particles and CAM particles generates voids, which result in a geometrically tortuous ion path and a high grain-boundary resistance at the SE/SE interfaces ($r_{i,gb}$). Moreover, the charge-transfer resistance at SE/CAM interface ($r_{i/e}$) also causes high resistance because of the chemical instability of the SE.¹¹ These microstructural factors (void space, tortuous ion path, grain-boundary, and SE/CAM interface) are generally considered as rate-limiting factors during SSE-based-ASSB operation;^{10,11} consequently, research

to resolve these issues and reduce the high resistance has drawn attention.^{10,12}

In general, the bottleneck in high-power-ASSB development has shifted from maximizing the ionic conductivity of SEs to designing composite-cathode microstructures for rapid charge-transport with maximizing CAM loading. Thus, it is crucial to understand how microstructural factors impact the charge-transport processes quantitatively. The major difficulty in analyzing the processes is the lack of an experimental methodology for quantifying the interfacial resistances caused by each microstructural factor. Consequently, composite-cathode resistances have been predominantly analyzed by electrochemical impedance spectroscopy (EIS) using conventional model circuits intended for the analysis of LE-based batteries (Figure 1a,b).^{6,8,13–18} However, the data generated by such LE based-circuits differ remarkably from the experimental results of SE-based composite cathodes.^{19,21}

These differences have provoked controversy. Several studies have suggested that the tortuous ion path is the major cause of the drop in ionic conductivity of SE-based

Received: January 23, 2024

Revised: April 23, 2024

Accepted: April 24, 2024

Published: May 13, 2024



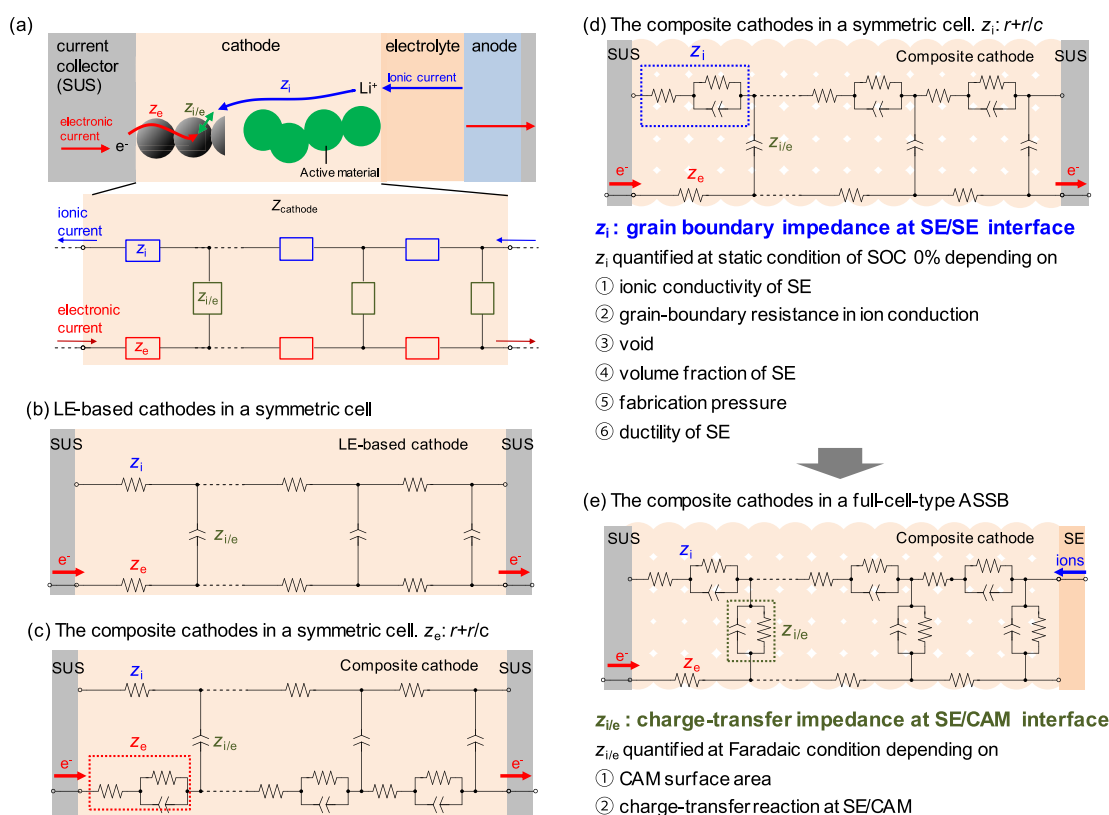


Figure 1. (a) Illustration of the electrochemical process in batteries (top) and schematic of the corresponding transmission-line-model (TLM)-based circuit (bottom). (b–e) TLM-based circuits for the (b) liquid-electrolyte-based cathode in a symmetric cell with electron–electron connections, (c, d) solid-electrolyte (SE)-based composite cathodes in a symmetric cell with electron–electron connections, and (e) SE-based composite cathode in full-cell-type all-solid-state batteries (ASSBs) with electron–ion connections. z_i in (b) represents $r_{i,bulk}$, whereas that in (c) and (d) corresponds to $r_{i,bulk} + r_{i,gb}/c_{i,gb}$; z_e in (c) represents $r_{i,bulk} + r_{i,gb}/c_{i,gb}$.

composite cathode materials compared to LEs.^{14,15} In contrast, other EIS studies have proposed that an additional resistance other than the tortuous ion path is the origin of high resistance.^{19–23} Siroma et al. proposed two different transmission line model (TLM)-based circuits featuring the additional resistance in an extra component—a parallel circuit containing a resistor and capacitor (R/C)—connected to z_e or z_i ¹⁹ (Figure 1c,d). Slight variations of this model have also been employed by Minnmann et al.²² and our group.²³ However, it is still uncertain, which circuit is most suitable for representing the composite-cathode microstructures and for simulating the charge-transport properties of composite cathodes. Consequently, there remains an unanswered question: what is the main mechanism resulting in the ionic conductivity of composite cathodes being significantly lower than those of LEs?

Therefore, in this study, the impact of the microstructural factors was systematically evaluated using model circuits (Figure 1d,e), having two key interfacial resistance components (r_{gb} and $r_{i/e}$). First, the validity of the model circuit (Figure 1d) was reassessed using symmetric cells. Compared to the circuit in Figure 1c,^{19,22} that in Figure 1d assuming the grain-boundary resistance ($r_{i,gb}$) component of ionic conduction was found to represent the composite-cathode microstructure under various conditions reasonably well, for example, with respect to the void, volume fraction of SE, grain boundaries, and fabrication pressure. These results clearly show that the low ionic conduction of the composite cathode is derived from a high $r_{i,gb}$ rather than the tortuous ion path.

Second, and most importantly, a circuit (Figure 1e) reflecting charge transfer at the SE/CAM interface was updated for full-cell-type ASSBs. This circuit enables the individual quantification of $r_{i,gb}$ and $r_{i/e}$. Based on the EIS analysis of ASSBs using the circuit in Figure 1e, we found that use of smaller CAM particles led to a higher cathodic resistance, even though it led to a decrease in $r_{i/e}$, resulting in a lower cathode resistance in LE-based batteries.^{24,25} The higher cathodic resistance was determined to be caused by the increase in $r_{i,gb}$, which is not observed in LE-based batteries.^{24,25} This result also suggests that there is a trade-off relationship between $r_{i,gb}$ and $r_{i/e}$ in ASSBs when unimodal CAM particles are used. Furthermore, we provide composite-cathode design strategy to help surpassing the undesired trade-off, demonstrating high-power and high-energy performance ASSBs.

RESULTS

Validity of the Model Circuits Representing Differences Between the Pristine SE and Composite Cathodes. Composite cathodes typically comprise an SE/CAM mixture and a small amount of an electron-conduction-enhancing carbon. Several charge-transport processes occur in the thick composite-cathode layer during ASSB operation as ions and electrons are transported through the SE and CAM, respectively, and stored in the CAM after charge-transfer reaction at the SE/CAM interface (Figure 1a). Although these charge-transport processes occur in a complex mixed-conducting state in thick composite cathodes, most composite cathodes are typically simulated using a simple flat-electrode

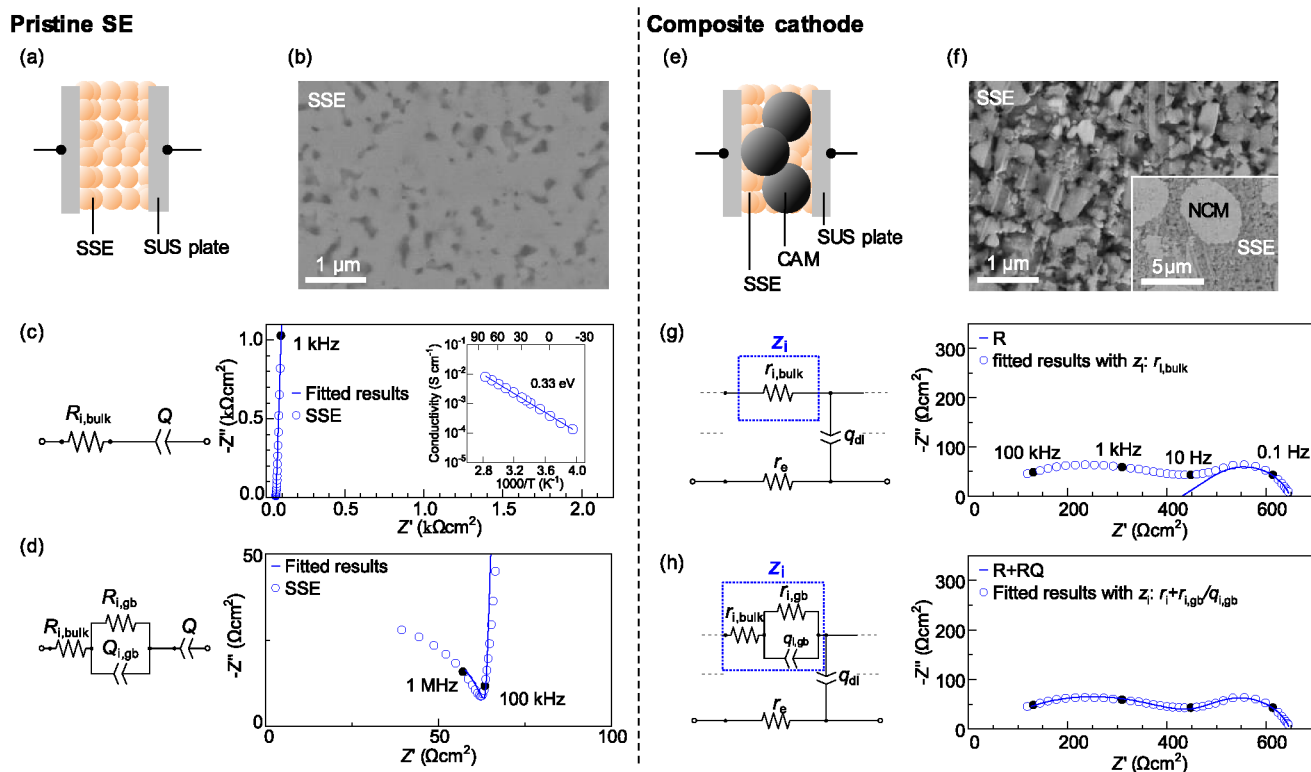


Figure 2. Schematics of symmetric cells containing a (a) pristine sulfide-based SE (SSE) and (e) composite cathode with electron–electron connections. Cross-sectional SEM images of (b) pristine SSE pellets and (f) composite cathode pellets pressed at 300 MPa (SSE:NCM = 50:50 v/v). Nyquist plots with fits obtained using model circuits for the (c, d) pristine SSE and (g, h) composite cathodes; the equivalent circuits correspond to (c) $R + Q$ and (d) $R + R/Q + Q$. z_i in the TLM-based circuits corresponds to (g) $r_{i,bulk}$ and (h) $r_{i,bulk} + r_{i,gb}/c_{i,gb}$.

model for LE-based batteries.^{4,8} Consequently, the ionic and electronic impedances have not been adequately quantified. Thus, we suggest a TLM (Figure 1a), otherwise known as de Levie model,¹⁶ to elucidate the charge-transport processes in the composite cathodes comprehensively; here, two lines run parallel to the depth direction of the cathode, and the ion- and electron-conducting impedance elements (z_i and z_e) are arranged in series along the upper and lower lines, respectively, and bridged by the charge-transfer impedance ($z_{i/e}$). We demonstrate the validity of the model circuit (Figure 1d) by carefully assessing its ability to simulate EIS profiles first and then assessing its ability to represent the structural features of the composite cathodes.

First, to assess the ability of the circuit (Figure 1d) to simulate EIS spectra, the symmetric cells of a composite cathode and a pristine SE were fabricated and electrochemically analyzed. To fabricate symmetric cells, a pristine argyrodite SSE ($\text{Li}_{5.75}\text{PS}_{6.75}\text{Cl}_{1.25}$; LPSC) or a 1:1 (by volume) mixture of an SSE and a Ni–Co–Mn-based layered oxide (NCM) was filled in a cylindrical case and sandwiched between cylindrical steel–use–stainless (SUS) electrodes (Figure 2–e).

For comparison, EIS of the pristine SSE was performed using a symmetric cell, exhibiting linear Nyquist data (Figure 2c) that were fit using an equivalent circuit with a purely resistive component, $R_{i,bulk}$, that is typical of high-ionic-conductivity SSEs and whose shape is similar to that of LE-based cell.⁶ Note that the Nyquist plots obtained via high-frequency analysis showed a small semicircle at high frequencies (>100 kHz; Figure 2d), indicating a small contribution from grain-boundary ion conduction, even in

the pristine SE; these data were fit to the circuit expressed as $R_{i,bulk} + R_{i,gb}/Q_{i,gb}$. The ionic resistivity r_i ($= R_{i,bulk}/t_{SE} + R_{i,gb}/t_{SE}$) was determined to be $809 \Omega\cdot\text{cm}$ ($r_{i,bulk} = 178 \Omega\cdot\text{cm}$, $r_{i,gb} = 631 \Omega\cdot\text{cm}$), where t_{SE} is the thickness of the SE layer.

The composite-cathode-based symmetric cell exhibited Nyquist plots with two semicircles in the high- and low-frequency regions (>10 and <10 Hz, respectively; Figure 2g), which were fit using two different TLM-based circuits (Figure 2g, h). A mathematical description of the TLM-based circuits (Figure 1b–d) is given by eq 1:^{19,20}

$$Z_{\text{cathode},e-e} = \frac{z_i z_e}{z_i + z_e} t_{\text{cathode}} + \frac{2Z_e^2 \sqrt{z_{i/e}}}{(z_i + z_e)^{3/2}} \cdot \frac{\cosh\left(t_{\text{cathode}} \sqrt{\frac{z_i + z_e}{z_{i/e}}}\right) - 1}{\sinh\left(t_{\text{cathode}} \sqrt{\frac{z_i + z_e}{z_{i/e}}}\right)} \quad (1)$$

Here, z_i and z_e [$\Omega\cdot\text{cm}$] represent the specific resistances of ions and electrons, respectively; $z_{i/e}$ [$\Omega\cdot\text{cm}^3$] represents the charge-transfer impedance per unit volume; and t_{cathode} [cm] is the cell thickness. Given that the state of charge (SOC) of CAMs in symmetric cells is zero, $z_{i/e}$ has been assumed to be purely capacitive.¹⁸

In Figure 2g, the purely resistive components $r_{i,bulk}$ and r_e were assigned to z_i ^{14,17–19,21,26} and z_e , respectively, assuming that there is only intragrain ion conduction in the composite cathode, as for the LE-based cell and pristine SE cell. However, the simulated profiles deviated considerably from the experimental spectra;¹⁹ briefly, teardrop-shaped plots with

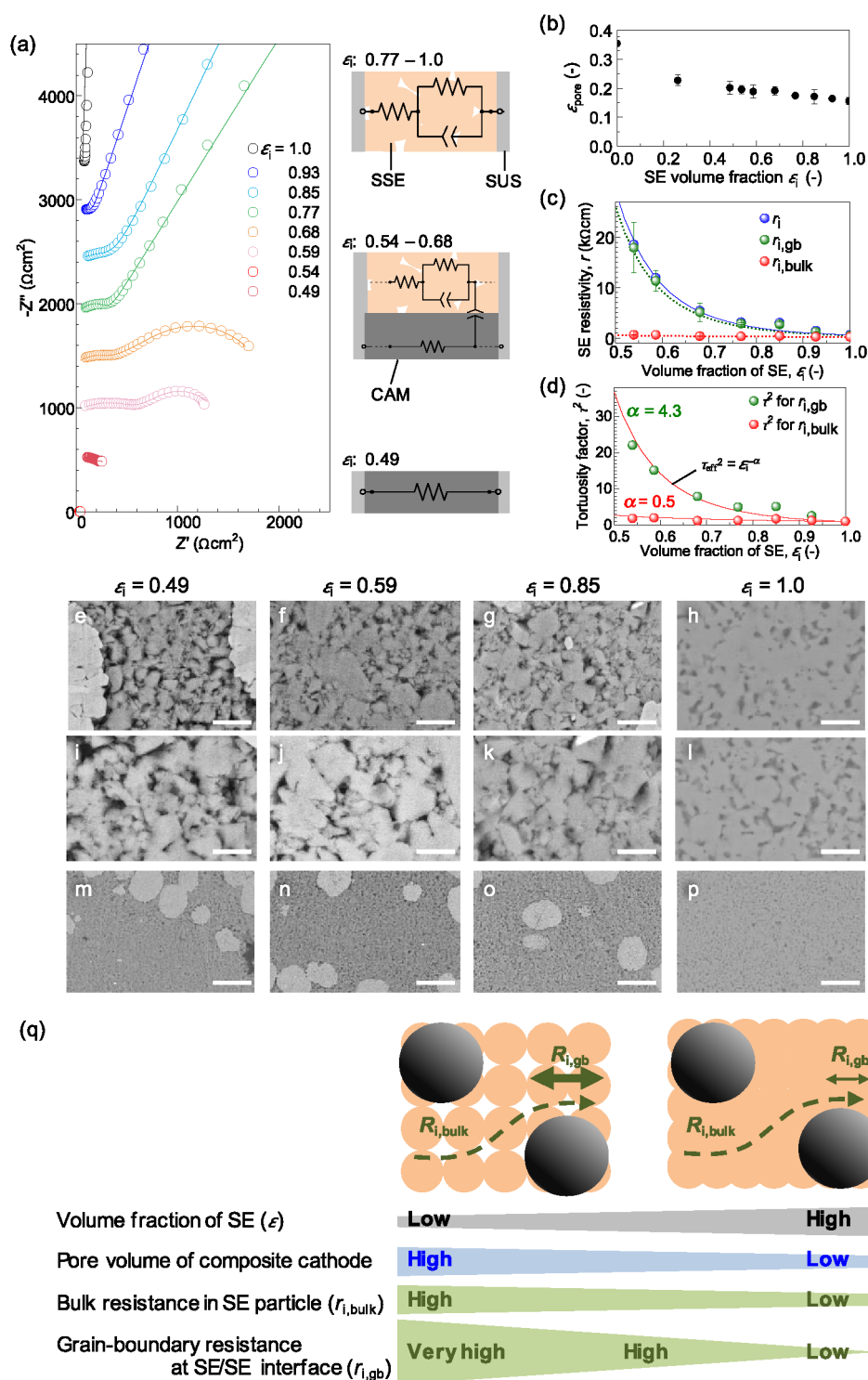


Figure 3. (a) Nyquist plots of the composite cathodes and the corresponding fits obtained using different model circuits for various SE volume fractions (ϵ_i). (b) Pore volume fraction of the composite cathodes. (c) Resistivity of the composite cathodes as a function of ϵ_i . (d) Tortuosity factor of the resistive components as a function of ϵ_i with fits obtained using the Bruggeman relationship (eq 5). Cross-sectional SEM images of the composite cathodes with ϵ_i values of (e, i, m) 0.49; (f, j, n) 0.59; (g, k, o) 0.85; and (h, l, p) 1.0. (scale bars: [e–h] 1 μ m, [i–l] 500 nm, and [m–p] 5 μ m). (q) Schematic of the relationship between volume fraction of SE and resistive components ($r_{i,bulk}$ and $r_{i,gb}$) in composite cathodes.

poor and acceptable fits were acquired at high and low frequencies, respectively. The ionic resistivity r_i ($=r_{i,bulk}$) was determined to be 12800 Ω -cm, which is approximately 16 $\times$ higher than the r_i ($=r_{i,bulk} + r_{i,gb}$) of the pristine SSE. Although the significantly higher ion-conduction resistivity of the composite cathode than that of the pristine SSE pellets has

been linked to the void-space-induced elongation of the tortuous ion path in previous studies,¹⁵ the increase in resistivity is much higher than that expected from such elongation.¹⁴

In contrast, in Figure 2h, ascribing the $r_{i,bulk}$ - and $r_{i,gb}/c_{i,gb}$ -component to z_i (Figure 1d) provided consistent fits to

impedance spectra in both the high- and low-frequency regions. The fit-related parameters $r_{i,bulk}$ and $r_{i,gb}$ (433 and 12700 $\Omega\cdot\text{cm}$) were 2- and 20-times higher than those of the pristine SSE cell, respectively; the reason for this is discussed later. These results demonstrate that the circuit (Figure 1d) yielded decent fits to the EIS data of composite cathodes (featuring two semicircles). In previous reports, a decent fit has also been shown using the circuit (Figure 1c).^{19,22} Thus, two questions remain: which circuit (Figure 1c,d) represents the physical properties of the composite-cathode microstructure best and what is the physical meaning of the extra circuits $r + r/c$?

Second, to assess the ability of the circuits to represent the structural features of the composite cathodes, cross-sectional SEM imaging of the symmetric cells (Figure 2b,f, respectively) was performed. The cross-sectional image of a pristine SSE pellet (Figure 2b) shows a monolithic morphology, indicating that the ductile particles had been well-sintered into a dense body under the applied pressure. In contrast, the composite cathode had voids and grains between SSE particles (Figure 2f). Therefore, the pristine SSE and the composite cathode differed considerably in microstructure, despite being similarly fabricated.

Bridging the apparent gap between the pristine SSE and composite cathodes, microstructural analyses and EIS were conducted with stepwise decrements in the SSE volume fraction of the composite cathode, ε_i . Wide-angle cross-sectional SEM images of the composite cathodes show homogeneous distributions of LPSC and NCM (Figure 3m–p). However, the magnified images clearly show the growth of void space and grain boundaries between LPSC particles with decreasing ε_i (Figures 3e–h and S3). The pore volume of void space in the pellets (ε_{pore}) was determined using the pellet volume (Figure 3b); ε_{pore} increased slightly with decreasing ε_i ($\varepsilon_{pore} = 0.16$ and 0.23 at $\varepsilon_i = 1$ and 0.26 , respectively). These results indicate that the addition of the bulky rigid NCM ($\varepsilon_i = 0$, $\varepsilon_{pore} = 0.35$) to SSE particles prevented the sintering of SSE particles even at high pressures, leading to the slight increase in pore volume.

EIS measurement of the composite cathode was performed with decreasing SE volume fraction ε_i . The Nyquist plots of the composite cathode obtained at ε_i values of 0.49–1.0 (Figure 3a) were fit using certain equivalent circuits (Figure 3a, right). First, at ε_i values of 0.77–1.0, where the electron-conduction resistivity (r_e) is much higher than that of r_i , linear plots with a small semicircle appeared at high frequencies (Figure 3a). This feature was fit using the curve defined by the $R_{i,bulk}$ - and $R_{i,gb}/Q_{i,gb}$ -containing circuit (Figure 3a; top-right panel), which is practically identical to the TLM-based z_e -free circuit with $z_i = r_{i,bulk} + r_{i,gb}/q_{i,gb}$. The semicircle diameter increased with decreasing ε_i , followed by an increase in $R_{i,gb}$, with almost constant $R_{i,bulk}$. Second, at ε_i values of 0.54–0.68, where the values of r_i and r_e are similar, the low-frequency plots show teardrop shape. The low-frequency limit of the real part of the impedance can be mathematically expressed by eq 2:¹⁹

$$\lim_{\omega \rightarrow 0} Z'_{cathode} = r_e t_{cathode} \quad (2)$$

The sudden change in the low-frequency impedance at $\varepsilon_i < 0.77$ can be attributed to the decrease in the resistivity in r_e , indicating that volume fraction of NCM reached the electrical percolation threshold.^{27,28} The plots in this ε_i range (0.54–0.68) could also be fit appropriately using the TLM circuit

shown in Figure 1d with $z_i = r_{i,bulk} + r_{i,gb}/q_{i,gb}$ and $z_e = r_e$ (Figure 3a; middle-right panel). Note that the plot corresponding to $\varepsilon_i = 0.54$ showed smaller semicircles than those of $\varepsilon_i = 0.59$ at both high and low frequencies owing to a lower r_e (see magnified plots in Figure S2). Third, at $\varepsilon_i = 0.49$, where r_e is lower than r_i , the EIS plots converged to a point on the real axis with $Z'' = 0$, impeding accurate fitting with the TLM-based circuit. Thus, the plots were fit using an R circuit representing pure resistance (Figure 3a; bottom-right panel), indicating that the electron-conducting path had been adequately established. Distinguishable semicircles were not obtained at $\varepsilon_i < 0.49$.

Given the above results, the EIS spectral analyses reveal that semicircles typical of TLM-based circuits with an extra r/c circuit appear when ion-conduction is dominant ($r_e > r_i$ at ε_i values of 0.54–1.0). In addition, microstructural analysis revealed that the high-frequency impedance increase is followed by growth of void and grain boundary of SSE particles. Moreover, the composite cathodes with ε_i values of 0.54–1.0 exhibited capacitance ($c_{i,gb}$) values of 10^{-11} to 10^{-8} F (Table S1), which can be attributed to grain-boundary ion-conduction capacitance.²⁹ It should be noted that the TLM-based circuits using z_e with $r_{e,bulk} + r_{e,gb}/q_{e,gb}$, for example, the circuit suggested by Siroma et al.¹⁹ and used by Minnmann et al.²² (z_e with $r_{e,bulk} + r_{e,gb}/q_{e,gb}$; Figure 1c) and the circuit used by our group (z_e with $r_{e,bulk} + r_{e,gb}/q_{e,gb}$ and z_i with $r_{i,bulk} + r_{i,gb}/q_{i,gb}$)²³ offer numerically good fits to the impedance spectra in Figures 2g and 3a. However, the parameters used for the fitting are physically meaningless in cases where the CAM is a good electronic conductor, for example, NCM, because it is unlikely that there is a high capacitance ($q_{e,gb}$) in a good electronic conductor. In fact, the capacitance of approximately 10^{-9} F obtained by EIS analysis (Table S1) is more likely attributable to the capacitance of grain-boundary ion-conduction.²⁹ Therefore, for composite cathodes, the circuit (Figure 1d) with $r_{i,bulk} + r_{i,gb}/q_{i,gb}$ ascribed to the ion-conduction-related impedance (z_i) is more reasonable than others.

Lastly, for a quantitative discussion of ion transport in composite cathodes, the individual resistive components (r_i , $r_{i,bulk}$, and $r_{i,gb}$) were quantified using the aforementioned circuits. The ε_i -dependent plots of the resistive components showed substantial differences (Figure 3c). At a low SE fraction, $\varepsilon_i (= 0.54)$, r_i increased significantly to a high resistivity of 18.4 k $\Omega\cdot\text{cm}$, highlighting the dominance of $r_{i,gb}$ (17.7 k $\Omega\cdot\text{cm}$), which is 29-times higher than that of $r_{i,bulk}$ (0.61 k $\Omega\cdot\text{cm}$), whereas $r_{i,bulk}$ and $r_{i,gb}$ were similar at a high SE volume fraction, $\varepsilon_i = 1.0$, ($r_{i,bulk} = 0.17$ and $r_{i,gb} = 0.63$ k $\Omega\cdot\text{cm}$).

The tortuosity factor, τ^2 , allows us to describe quantitatively the obstruction of ion transport in individual path of composite cathodes, as given by eq 3 and eq 4:^{30,35}

$$\sigma_{i,eff} = \varepsilon_i / \tau_{eff}^2 \times \sigma_{i,0} \quad (3)$$

$$\sigma_i = 1/r_i \quad (4)$$

Here, $\sigma_{i,eff}$, $\sigma_{i,0}$, and τ_{eff} represent the effective ionic conductivity of the composite cathode, ionic conductivity of pristine SE pellets, and effective tortuosity, respectively. Of note, τ_{eff} differs from the geometrical tortuosity, τ , which is defined as the ratio between the length of a curved conducting path and the shortest start-to-end point distance. Thus, obstructive effects other than the elongation of the ion-conducting path could lead to a higher τ_{eff} than τ . The tortuosity factors (τ_{eff}^2) of the composite cathode with the low

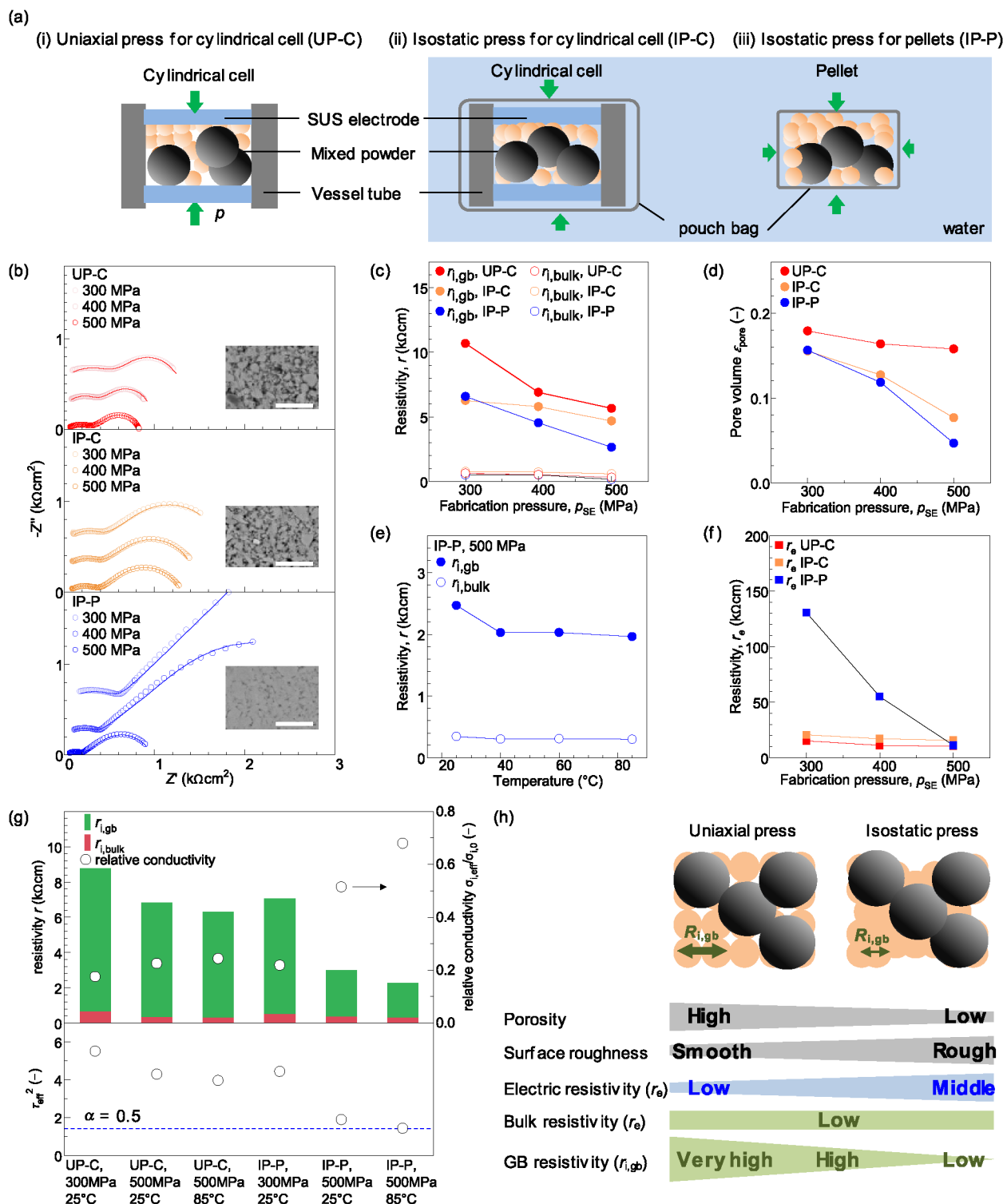


Figure 4. (a) Schematic of the different pressurization methods used for fabricating the composite cathode with an ε_i value of 0.58: (i) uniaxial pressing of a cylindrical cell (UP-C), (ii) isostatic pressing of a cylindrical cell (IP-C), and (iii) isostatic pressing of cathode pellets in a pouch bag (IP-P). (b) Nyquist plots of the composite cathodes fabricated by UP-C, IP-C, and IP-P; inset: cross-sectional SEM images of the composite cathodes. (c) Bulk resistivity ($r_{i,bulk}$) and grain-boundary resistivity ($r_{i,gb}$) determined using the fits shown in (b). (d) Pore volume fraction of the composite cathodes as a function of the fabrication pressure. (e) Resistivity of the composite cathodes fabricated at 500 MPa as a function of temperature. (f) Electronic resistivity of the composite cathodes (r_e) determined using the fits shown in (b). (g) Ionic resistivity and conductivity of the composite cells (top), as well as the corresponding effective tortuosity (bottom); the blue dashed line represents the tortuosity factor (τ_{eff}^2) estimated from the ideal Bruggeman relationship for spherical particles with $\alpha = 0.5$. (h) Bulking properties and resistive components of the composite cathodes pressed with the uniaxial press and the isostatic press.

SE volume fraction ($\varepsilon_i = 0.54$) were 1.9 and 22 for $r_{i,bulk}$ and $r_{i,gb}$, respectively (Figure 3d). Considering that tortuosity

factors of liquid electrolytes below 10 have been reported for porous cathodes at electrolyte volume fractions >0.3 ,^{31–33} the

tortuosity factor of $r_{i,gb}$ (22) is unexpectedly high compared to those of LE-based cathodes.

Here, the relationship between the tortuosity factors and the SE volume fraction can be interpreted with the empirical Bruggeman relationship shown in eq 5:^{34,35}

$$\tau_{\text{eff}}^2 = \varepsilon_i^{-\alpha} \quad (5)$$

The original relationship was developed to describe the geometrical tortuosity of spherical insulating particles. The Bruggeman exponent α is 0.5 in an ideal Bruggeman-type relationship for spherical particles³⁶ but higher for more complex situations. After fitting (Figure 3c), the α for $r_{i,bulk}$ (0.52) was found to be similar to that of the ideal Bruggeman-type relationship, whereas that for $r_{i,gb}$ (4.3) was considerably higher. This suggests that the decrease in ε_i induces a slight increase in $r_{i,bulk}$ via the elongation of the ion-conducting path. However, the significant increase in $r_{i,gb}$ surpasses the ideal Bruggeman scenario. Overall, it can be concluded that the significant increase r_i values with low SE loading is primarily attributed to grain-boundary growth rather than the elongation of the tortuous ion path (Figure 3q).

Impact of Fabrication Pressure on Ion-Conducting Processes. The fabrication of composite cathodes, particularly pressurization, affects their microstructures and charge-transport processes.^{26,37} Our developed circuit (Figure 1d) can be employed to quantify these. To investigate the impact of fabrication pressure, the composite cathodes were prepared using three different pressurization processes (Figure 4a): (i) filling the NCM-LPSC powder mixture in a cylindrical cell and then pressing with two SUS electrodes using a conventional uniaxial press;^{8,19,21–23} (ii) pressurizing the powder mixture in a cylindrical cell with two SUS electrodes using an isostatic press with fluidic media; and (iii) laminating pellets of the powder mixture with a flexible pouch bag and then pressurizing them using an isostatic press.¹ These methods are denoted UP-C, IP-C, and IP-P, respectively.

Nyquist plots of the symmetric cells with the differently fabricated composite cathodes showed two semicircles (Figure 4b), whose diameter decreased with increasing pressure, indicating decreases in both r_i and r_e . For a quantitative analysis, the $r_{i,bulk}$ and $r_{i,gb}$ determined using the circuit shown in Figure 1d were plotted against the fabrication pressure (Figure 4c). In contrast to that of $r_{i,bulk}$, the decrease of $r_{i,gb}$ is remarkable with increasing fabrication pressure for all the cells. The $r_{i,gb}$ of the IP-P cell decreased considerably, reaching 2650 $\Omega \cdot \text{cm}$ at 500 MPa, whereas that of the IP-C cell was 4680 $\Omega \cdot \text{cm}$. The cross-sectional SEM image of the IP-P cell (Figure 4b, inset bottom-right) showed that the SSE particles densified adequately (see Figure S4 for details). The pore volume of the IP-P cell was reduced ($\varepsilon_{\text{pore}} = 0.046$) at 500 MPa (Figure 4d). This indicates that the pressurization through fluidic-media (water) using a pouch bag enabled the smooth sintering of SSE particles in the composite cathode. This compression was smoother at temperatures over 40 °C (Figure 4e). The effects of the fabrication pressure on r_e were also shown (Figure 4f). At 500 MPa, the r_e of all the cells decreased to below 15 $\text{k}\Omega \cdot \text{cm}$, whereas at 300 MPa, the r_e values of the cells using SUS electrodes (UP-C and IP-C) were considerably lower than that of the cell using a pouch bag (IP-P), implying that the pressurization using the SUS electrode led to stress being preferentially imposed on the CAM particles.

The overall impact of pressurization on the investigated cells is summarized in Figure 4g, which shows $r_{i,bulk}$, $r_{i,gb}$, the relative

conductivity $\sigma_{i,\text{eff}}/\sigma_{i,0}$, and τ_{eff}^2 determined using eq 3. The r_i value of the IP-P cell subjected to 500 MPa at 85 °C (2260 $\Omega \cdot \text{cm}$) yielded a τ_{eff}^2 of 1.46, which is close to that obtained under ideal Bruggeman-type conditions (1.41 at $\alpha = 0.5$). This implies that the fluidic-medium-assisted pressurization allows the intense densification of SE particles with a low Young's modulus to achieve a pore volume level comparable to that of the pristine SE pellets in the isostatic press. On the other hand, in the uniaxial press, the hard SUS plate pressurizes SSE and CAM evenly, resulting in less deformation of the SSE in the composite cathode compared to that of isostatic press (Figure 4h).

Impact of CAM Particle Size on Ion Conduction at Interfaces: Quantification of Charge-Transfer Resistance $r_{i/e}$ by In Situ Impedance Spectroscopy of ASSBs. Charge transfer at the SSE/CAM interface is critical because it directly impacts the ASSB performance as a result of insufficient physical contact, chemical reactions, electrochemical reactions, and space-charge layers;¹⁰ these factors can be assessed by quantifying the charge-transfer impedance. However, EIS analyses of composite cathodes have not been interpreted well because of spectral overlap of the grain-boundary and charge-transfer impedance components. However, parameters representing the grain-boundary impedance of the composite cathode have been successfully extracted using the proposed circuit for symmetric cells (Figure 1d), as reported above; therefore, an extended circuit for full-cell-type ASSBs (Figure 1e) was subsequently developed to quantify the charge-transfer impedance of the composite cathode.

To extract the charge-transfer impedance, a suitable circuit must be selected based on the cell configuration, components, and the requirement to evaluate the charge-transport-related parameters. In situ impedance spectroscopy of a full-cell-type ASSB containing the composite cathode was performed using a model that incorporated the circuit in Figure 1e (Figure 5). In

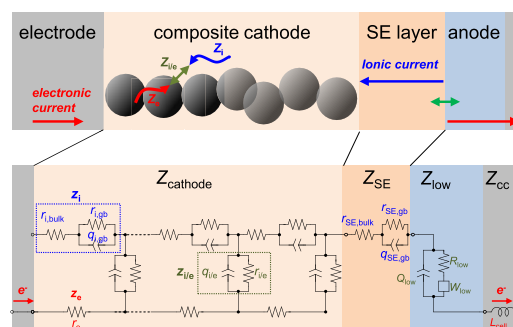


Figure 5. Model circuit for full-cell-type ASSBs.

the TLM-based composite-cathode circuit, a component of the parallel circuit containing $r_{i/e}/q_{i/e}$ was assigned to $z_{i/e}$, which represents charge transfer at the SE/CAM interface. The total impedance of the full-cell-type ASSB is expressed by eq 6:

$$Z_{\text{total}} = Z_{\text{cathode},i-e} + Z_{\text{SE}} + Z_{\text{low}} + Z_{\text{CC}} \quad (6)$$

EIS profiles representing diffusion in both the anode and cathode of the full-cell-type ASSB, which occurred at low frequencies (<30 Hz), were fit using a circuit with a Warburg element in the Z_{low} region. The accurate quantification of the individual diffusion constants requires separate EIS measurements of the anode and cathode using a three-electrode cell with a circuit such as that in Figure S12, which will be

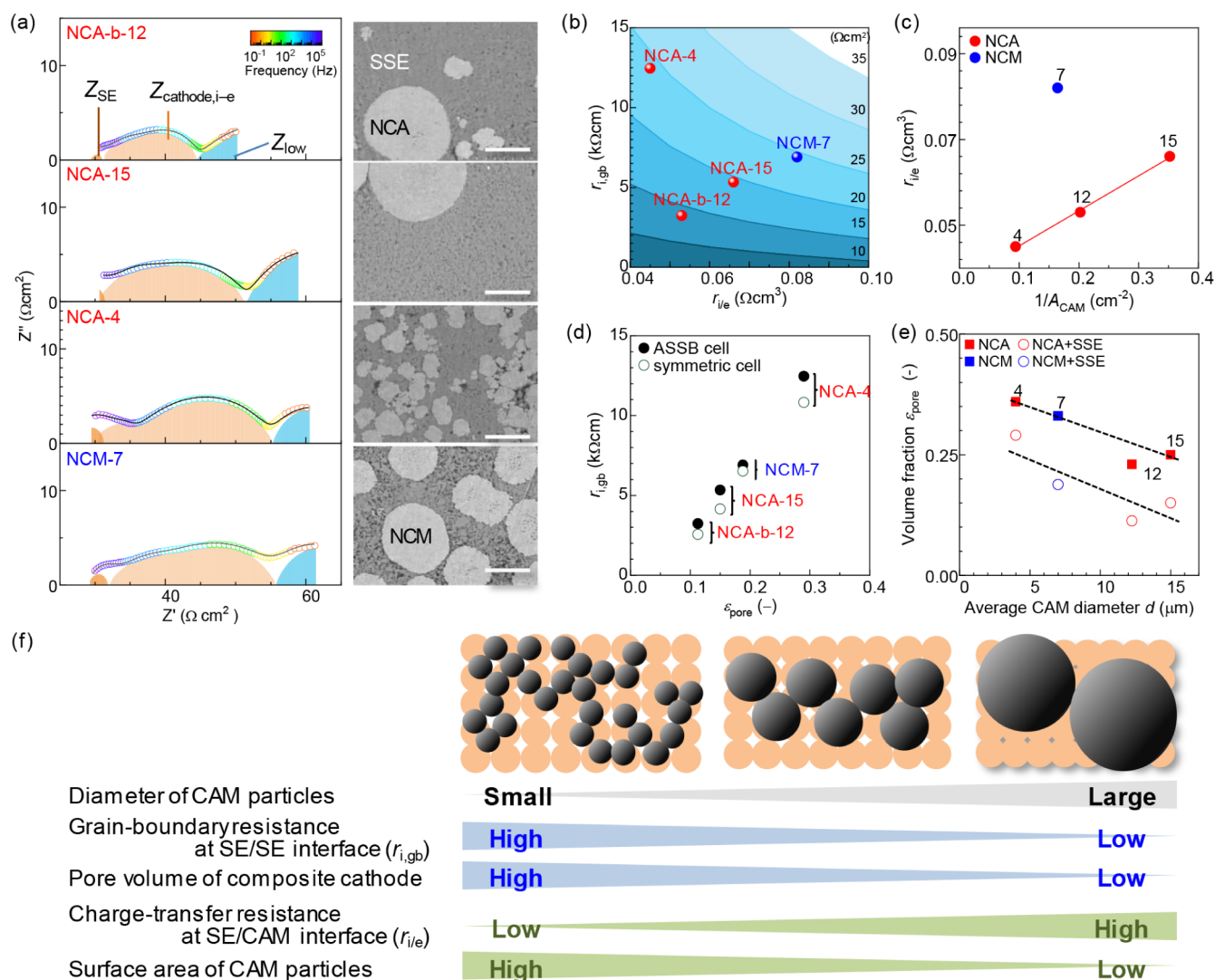


Figure 6. (a) Nyquist plots of ASSBs with the In/SSE/SSE + CAM + CF configuration containing NCA-b-12, NCA-15, NCA-4, or NCM-7 as the CAM; cross-sectional SEM images of the composite cathodes are shown on the right (scale bars: 5 μm). The y-axis scale is magnified by 1.3-times compared with that of the x-axis. Color-filled semicircles are visual representations of the different impedance elements, as determined by the fitting results. (b) Contour map of the cathodic resistance as a function of $r_{i,gb}$ and $r_{i/e}$, with overlaid coordinates of the ASSBs. (c) $r_{i/e}$ plotted against the inverse of the CAM surface area. (d) $r_{i,gb}$ plotted against the pore volume fraction. (e) Pore volume fraction as a function of the CAM particle diameter. (f) Bulking properties and interfacial resistances of the composite cathodes with unimodal CAM particles.

considered in a future study. The cathodic impedance in the TLM-based circuit with ion–electron connections is expressed by eq 7:²⁰

$$Z_{cathode,i-e} = \frac{z_i z_e}{z_i + z_e} t_{cathode} + \frac{\sqrt{z_{i/e}}}{(z_i + z_e)^{3/2}} \cdot \frac{(z_i^2 + z_e^2) \cosh\left(t_{cathode} \sqrt{\frac{z_i + z_e}{z_{i/e}}}\right) + 2z_i z_e}{\sinh\left(t_{cathode} \sqrt{\frac{z_i + z_e}{z_{i/e}}}\right)} \quad (7)$$

The cathodic impedance of ASSBs is significantly affected by the CAM particle size and composition.^{38–41} However, the impact of experimental conditions on the interfacial impedances has not been systematically investigated. Thus, the interfacial impedances of ASSBs were investigated using three types of unimodal CAM particles having various particle

diameters and compositions: small- and large-diameter a Ni–Co–Al-based layered oxide (NCA) particles (4- μm -sized NCA-4 and 15- μm -sized NCA-15, respectively) and medium-diameter NCM particles (7- μm -sized NCM-7).

The impedance spectra of full-cell-type ASSBs having the SUS/In/SSE/SSE + CAM + CF/SUS configuration and their model-circuit (see Figure 5)-based fits are shown in Figure 6a; the parameters used for the fitting listed in Table S2. By using values determined from the EIS profiles of the composite-cathode-based symmetric cell as the initial values, z_i and $z_{i/e}$ could be separated well. Fortunately, the capacitances of z_i and $z_{i/e}$ were sufficiently different to enable separation. Even though all spectra showed various shapes depending on the particle diameter and chemical composition, they were adequately fit using the model circuit.

The EIS profile of the cells can be separated into high-, mid-, and low frequency spectra. The high- and low-frequency spectra of all the cells (>1 kHz and <30 Hz, respectively) were fit appropriately using almost identical parameters. The high-

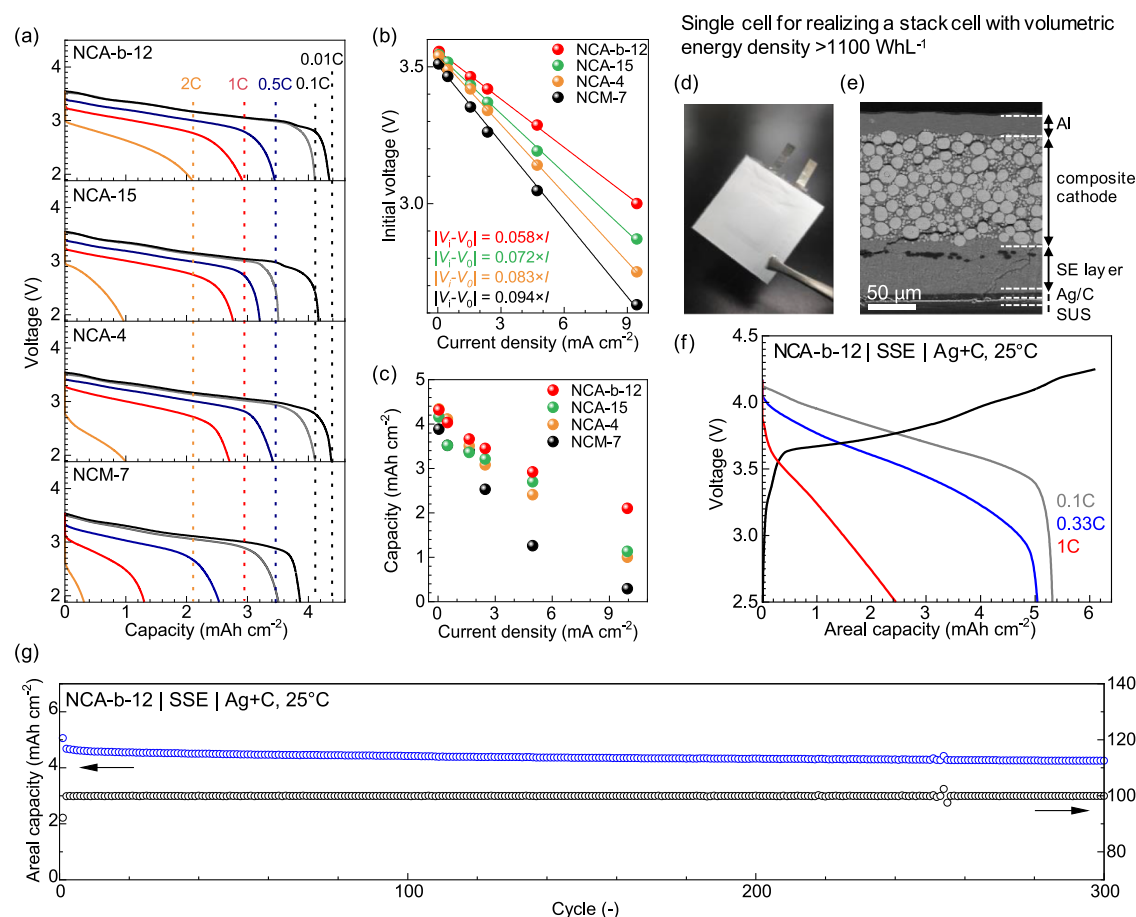


Figure 7. (a) Discharge profiles acquired at 0.1–2 C (0.5–10 mA·cm⁻²; projected cathode capacity = 5 mA·h·cm⁻²) for cylindrical cells containing NCA-b-12, NCA-15, NCA-4, or NCM-7 in the In/SSE/composite cathode configuration. (b) Initial voltage and (c) discharge capacity of the cells containing NCA-b-12, NCA-15, NCA-4, or NCM-7 at various current densities. (d) Photograph of a pouch-type cell with the Ag–C/SSE/NCA-b-12 configuration (areal capacity = 6.3 mA·h·cm⁻²; 1.0 C = 6.3 mA·h·cm⁻²), and its (e) cross-sectional SEM image. (f) Rate capability of the single cell at 25 °C. (g) Cycling performance and Coulombic efficiency of the single cell at 25 °C in constant-current mode with charge/discharge rates of 0.1 C/0.33 C (voltage window, 2.5–4.25 V).

frequency spectra, which corresponded to the Z_{SE} element (Figure 6a; orange region) representing the bulk and grain-boundary resistances of the SE separator ($r_{SE} = 32.6 \pm 1.4 \Omega \cdot \text{cm}^2$). Moreover, the low-frequency spectra (Figure 6a; blue region), which corresponds to the Z_{low} element representing charge diffusion in the anode and cathode, were fit appropriately using the parameter R_{low} ($13.9 \pm 2.6 \Omega \cdot \text{cm}^2$). The midfrequency spectra, which were attributed to the cathodic impedance $Z_{cathode,i-e}$ (Figure 6a; pale-orange region), showed various shapes because of the overlap between the two semicircles at high and low frequencies. The cathodic resistance $R_{cathode}$ determined using the real part of $Z_{cathode}$ increased as follows: NCA-15 < NCA-4 < NCM-7 (20, 24, and 25 $\Omega \cdot \text{cm}^2$, respectively); this indicates that the cathodic resistances depended significantly on the cathode composition. To interpret the spectral shape of the cathodic impedances, EIS simulations using eq 7 were used, providing clear insights into the resistive components ($r_{i,gb}$ and $r_{i/e}$) (Figure S5). Interestingly, increasing $r_{i/e}$ resulted in an increase in the high- and low frequency semicircles, whereas increasing $r_{i,gb}$ only affected the low-frequency semicircle. When the $r_{i,gb}$ is lower than $r_{i/e}$, the cathodic impedance spectrum form a teardrop shape (Figure S5e, left). Furthermore, as the variables related to the grain boundaries in ionic conduction ($r_{i,gb}$, $q_{i,gb}$) become higher, the teardrop-shaped spectrum first became a collapsed

semicircle (Figure S5e, center) and then changed to a double-peak form (Figure S5e, right). The semicircles of the double-peak form cannot be assigned to any of the resistive components because the formula for the cathodic impedance (eq 7) consists of the sum of the functions of two variables ($r_{i,gb}$ and $r_{i/e}$) instead of the functions of a variable ($r_{i,gb}$ or $r_{i/e}$). The EIS data for NCM-7, NCA-15, and NCA-4 have teardrop, collapsed-semicircle, and double-peak spectrum, respectively (Figure 6a).

To understand the origin of the different cathodic resistances, the relationships between $R_{cathode}$ and the interfacial resistive components ($r_{i,gb}$ and $r_{i/e}$) were reordered (Figure 6b). Because $R_{cathode}$ is a function of $r_{i,gb}$ and $r_{i/e}$ (eq 7), it can be expressed as a contour line in a two-dimensional map of $r_{i,gb}$ and $r_{i/e}$ (Figure 6b), and the ($r_{i/e}$, $r_{i,gb}$) coordinates of the three cells are plotted in Figure 6b. For example, the coordinates of NCA-4 and NCM-7 are located in the high-cathodic-resistance region (approximately 25 $\Omega \cdot \text{cm}^2$); however, their positions were opposed to those of $r_{i,gb}$ and $r_{i/e}$ (high $r_{i,gb}$ and low $r_{i/e}$ for NCA-4; low $r_{i,gb}$ and high $r_{i/e}$ for NCM-7); this indicates that the interfacial impedances differed significantly, even at similar level of cathodic resistances. Moreover, a comparison of plots of the small and large NCA particles (NCA-4 and NCA-15) revealed a notable trade-off between $r_{i/e}$ and $r_{i,gb}$. The decreasing CAM diameter led to a

lower $r_{i/e}$ but a higher $r_{i,gb}$, resulting in the higher cathodic resistance with smaller NCA particles than that of larger NCA particles. This trend has not been observed in LE-based batteries to the best of our knowledge.⁴¹

Next, we considered why the resistive components ($r_{i,gb}$ and $r_{i/e}$) show a trade-off relationship with the cathode diameter. This answer must be related to the microstructure. First, the charge-transfer resistance ($r_{i/e}$) plotted against the surface area of the NCA particles (A_{CAM} ; Figure 6c) shows that $r_{i/e}$ was proportional to $1/A_{CAM}$, indicating that the higher surface area of CAM particles led to the lower $r_{i/e}$. The slope of the regression line was presumed to represent the material property that characterizes charge-transfer reaction at the SSE/CAM interface, which in principle may vary with CAM type, SE composition, state of charge, and sample history; a value of $0.082 \Omega \cdot \text{cm}^5$ was obtained for the NCA particles. Moreover, the $r_{i/e}$ value of the NCM-7 was much higher than that of the NCA particles, indicating that charge transfer at the LPSC/NCM interface is slower than that at the LPSC/NCA interface. These results are similar to those of earlier studies for LE-based batteries comparing NCA and NCM.^{42,43} Second, the grain-boundary resistances, $r_{i,gb}$, plotted against the pore volume (Figure 6d) revealed almost identical increases in $r_{i,gb}$ with increasing pore volume for both cells; the $r_{i,gb}$ values for the full cells were slightly higher than those of the symmetric cells probably because of the carbon additive. Furthermore, a plot of the pore volume of tapped CAM particles against the CAM diameter (Figure 6e) revealed an interesting inverse correlation; that is, decreasing the CAM diameter leads to an increase in the pore volume. The most likely explanation of this phenomenon is that the smaller particles aggregate in a nonuniform fashion, unlike the larger particles, yielding more porous pellets. This phenomenon has been observed in powder engineering studies and known as the Roller's relationship.^{44,45} Furthermore, the trend is similar to that of the composite cathode pellets (Figure 6e). The cross-sectional SEM image of the NCA-4-based cathode (Figure 6a) showed CAM particle aggregates separated by micron-sized pores; however, the NCA-15-based composite cathodes had fewer pores (see detailed cross-sectional SEM images in Figure S6). Therefore, as summarized in Figure 6f, the use of smaller CAM particles led to a lower $r_{i/e}$ owing to their higher surface area but resulted in a considerably higher $r_{i,gb}$ owing to their higher porosity, causing a trade-off relationship between $r_{i/e}$ and $r_{i,gb}$ in the unimodal CAM system.

Next, a bimodal NCA-4/NCA-15 mixture having an average particle diameter of $12 \mu\text{m}$ (NCA-*b*-12) was introduced. The NCA-*b*-12-based composite cathode achieved a considerably lower pore volume (0.11) than the other cathodes (Figure 6e), suggesting that the infiltration of the small NCA-4 particles between the large NCA-15 particles helped densify the CAM particles; this led to the NCA-*b*-12-based composite cathode exhibiting the lowest R_{cathode} value ($14 \Omega \cdot \text{cm}^2$) owing to the lowest $r_{i,gb}$ ($3240 \Omega \cdot \text{cm}$) among the samples (Figure 6b). As a result, the use of bimodal CAM particles allows us to overcome the trade-off relationship between $r_{i/e}$ and $r_{i,gb}$ (Figure 6f). Therefore, in terms of selecting cathode materials, CAM particles that can maximize the surface area and minimize the porosity of composite cathodes should be used to reduce the cathodic resistance.

ASSB Cell Performance. Electrochemical tests of the ASSBs were performed at 25°C using thick composite cathodes with NCA of 200 mAhg^{-1} . The ASSB was designed

to achieve energy densities $>800 \text{ W} \cdot \text{h} \cdot \text{L}^{-1}$ and required a single-cell areal capacity $>5 \text{ mA} \cdot \text{h} \cdot \text{cm}^{-2}$ in a stacked configuration (Figure S7 and Table S3). The performance of different ASSBs containing the four composite cathodes with a projected cathode capacity of $5 \text{ mAh} \cdot \text{cm}^{-2}$ was tested from low to high rates (0.01C–2C). The discharge profiles of the NCA-*b*-12-based cell revealed a slightly higher capacity of $4.1 \text{ mAh} \cdot \text{cm}^{-2}$ at a low discharge rate (0.1C; $0.5 \text{ mA} \cdot \text{cm}^{-2}$; Figure 7a, top) than that of the other cathodes (e.g., capacity of NCM-7-based cell = $3.5 \text{ mA} \cdot \text{h} \cdot \text{cm}^{-2}$; Figure 7a, bottom); however, the discharge profiles acquired at a high discharge rate (2C; $10 \text{ mA} \cdot \text{cm}^{-2}$) showed a significant difference. Notably, the NCA-*b*-12-based cell exhibited a discharge capacity 7-times higher than that of its NCM-7-based counterpart (2.1 and $0.3 \text{ mA} \cdot \text{h} \cdot \text{cm}^{-2}$, respectively) as a result of the considerably lower *IR* drop of the former than that of the latter (Figure 7b), leading to a higher initial discharge voltage (3.00 V) than that of the NCM-7-based cell (2.63 V). These results imply that the cathodic resistance directly affected the discharge capacity of the ASSB, particularly at high discharge rates (Figure 7c). The internal resistance of the NCA-*b*-12-based cell determined from the slope of the voltage drop $|V_i - V_0|$ ($58 \Omega \cdot \text{cm}^2$) was almost identical to that determined via EIS analysis ($R_{\text{SE}} + R_{\text{cathode}} + R_{\text{anode}} = 57 \Omega \cdot \text{cm}^2$).

Finally, a pouch-type cell of anode-free¹ ASSB was fabricated (Figure 7d) using a $105\text{-}\mu\text{m}$ -thick NCA-*b*-12-based composite cathode (projected areal capacity = $6.3 \text{ mA} \cdot \text{h} \cdot \text{cm}^{-2}$; Figure 7e) to achieve energy densities $>1100 \text{ W} \cdot \text{h} \cdot \text{L}^{-1}$ (Figure S6). The cell exhibited a discharge capacity of $5.3 \text{ mA} \cdot \text{h} \cdot \text{cm}^{-2}$ ($171 \text{ mAh} \cdot \text{g}^{-1}$) at 0.1C (Figure 7f). Importantly, a discharge capacity of $5.0 \text{ mA} \cdot \text{h} \cdot \text{cm}^{-2}$ ($159 \text{ mAh} \cdot \text{g}^{-1}$) was maintained at a high current density of $2.1 \text{ mA} \cdot \text{h} \cdot \text{cm}^{-2}$ (0.33C) without an appreciable decline in capacity ($135 \text{ mAh} \cdot \text{g}^{-1}$; 85% after 300 cycles; Coulombic efficiency = 99.9%; Figure 7g).

CONCLUSION

Charge-transport processes in the composite cathodes of ASSBs were described with a TLM-based circuit having two key interfacial parameters: the grain-boundary and charge-transfer impedances at the SE/SE and SE/CAM interfaces, respectively. The methodology allowed us to obtain a quantitative understanding of the impact of the microstructural factors of the composite cathodes (voids, tortuous ion path, grain boundaries, and SE/CAM interface), revealing that the low ionic conductivity of the composite cathode compared to that of LE-based batteries is attributed mainly to a high grain-boundary resistance rather than elongation of tortuous ion path. In addition, it was found that a lower SE fraction, lower fabrication pressure, and smaller CAM particles result in a higher grain-boundary resistance. Furthermore, we have proposed design strategies containing the use of bimodal CAM particles, which reduced the cathodic resistances as a result of the decrease of both $r_{i,gb}$ and $r_{i/e}$, demonstrating the operation of high-energy-density ASSB at high current densities.

The methodology reported in this paper can help effectively understand ASSB behavior and can act as a guide for exploring new SEs, modifying the SE/CAM interface, conducting deterioration-related mechanistic analyses, and integrating the ASSB components. In the future, further investigation should be conducted to develop materials for ASSBs with EIS methodologies based on the model circuit suggested in this

paper or on an improved model considering diffusion process in the anode and cathode.

MATERIALS AND METHODS

Material Preparation. An argyrodite-type SSE ($\text{Li}_{5.75}\text{PS}_{6.75}\text{Cl}_{1.25}$; LPSC; 99 wt %, 3 μm ; Mitsui Mining and Smelting), $\text{LiNi}_{0.90}\text{Co}_{0.05}\text{Mn}_{0.05}\text{O}_2$ (NCM; JX Nippon Mining and Metals), $\text{LiNi}_{0.94}\text{Co}_{0.04}\text{Al}_{0.02}\text{O}_2$ (NCA; Samsung SDI), and carbon fiber (CF, Teijin) were used. The crystal densities of LPSC, NCM, and NCA were 1.6, 4.69, and 4.65 $\text{g}\cdot\text{cm}^{-3}$, respectively. Before use, all the CAMs were coated with Li_2ZrO_3 using a previously reported sol–gel method.^{1,21}

Preparation of Composite Cathodes for the Symmetric Cell.

To conduct the EIS analysis of symmetric cells with electron–electron contact, composite cathodes were prepared by blending the SSE and CAM in a blade mixer. LPSC and NCM or NCA were used as the SSE and CAM, respectively. The volume fraction of the SSE in the composite cathode was set to 0.49–1.0. The cathode mixture was pelletized by setting the total volume of the mixture to 0.094 cm^{-3} ($V_{\text{cathode}} = m_{\text{SSE}}/d_{\text{SSE}} + m_{\text{CAM}}/d_{\text{CAM}}$), where m and d denote the mass and density of the material, respectively. Three types of pressurizing methods were employed. The first method involved uniaxial pressing (Model 4350, Carver) with a cylindrical pelletizing die (diameter, 13 mm). Here, the cathode mixture was filled into the die and then pressed at a specific pressure for 5 min. The second method involved isostatic pressing (EWIP-90-200, Ilshin Autoclave) with a die. Here, the cathode mixture was filled into the die, which was then vacuum sealed in a laminate bag and pressurized by the isostatic press. The third method involved isostatically pressing the composite cathode pellets. Here, the cathode mixture was filled into the die and preliminarily pelletized at 200 MPa. The preformed pellet was vacuum sealed in a laminate bag and pressurized by isostatic pressing. The fabrication pressure was set to 300–500 MPa, which yielded approximately 800- μm -thick pellets. Symmetric cells were prepared by Pt-sputter-coating the pellets on both sides and then placing them between SUS plates in a cylindrical case. A stacking pressure was applied to the cells by fastening the cap with a torque of 4 N·m. These procedures were all performed under dry conditions at a dew point below $-50\text{ }^{\circ}\text{C}$.

Fabrication of Cylindrical ASSBs. A cylinder-type ASSB with an areal capacity of 5 $\text{mA}\cdot\text{h}\cdot\text{cm}^{-2}$ was fabricated in a layered manner using the composite cathode, LPSC, and indium foil as the cathode, SE separating layer, and anode, respectively. The composite cathode was prepared by blending LPSC, CAM, and CF in a blade mixer (33:66:1 by weight). The three components of the cell—indium foil (100 μm), SSE powder (100 mg), and composite cathode powder (50 mg)—were added to a 13 mm-diameter cylindrical case and subjected to uniaxial pressing at 500 MPa for 5 min. The obtained cell was subjected to a stacking pressure by fastening the cap with a torque of 4 N·m. All the preparations were performed under dry conditions at a dew point below $-50\text{ }^{\circ}\text{C}$.

Fabrication of a Pouch-Type High-Energy ASSB. A pouch-type cell with a capacity of 6.3 $\text{mA}\cdot\text{h}\cdot\text{cm}^{-2}$ (composite cathode area = 2.89 cm^2) was fabricated. To prepare the anode, silver nanoparticle powder and carbon black powder (60 and 35 nm in diameter, respectively) were mixed (1:3 by weight) in *N*-methylpyrrolidone (NMP; 3 mL) containing 7 wt % polyvinylidene fluoride. After the mixture had been stirred using a planetary mixer (Thinky Corporation, AR-100) with zirconia balls at 1000 rpm for 3 min, NMP (3 mL) was added dropwise, and the resulting mixture was further stirred for a total of 300 min (90 min cooling followed by 3 min of ball milling to prevent overheating). The resulting slurry was coated onto a 10- μm -thick SUS foil using a 50- μm blade coater and dried at 80 $^{\circ}\text{C}$ for 20 min. The obtained electrode was further dried under vacuum at 100 $^{\circ}\text{C}$ for 12 h, and the thickness of the Ag–C layer was estimated to be 5 μm . The SE separating layer was prepared by wet coating. To that end, LPSC (98.5 wt %) and a nonaqueous acrylate binder (1.5 wt %) in xylene were blended in a planetary mixer at 2000 rpm for 6 min. The resulting mixture was coated on a

polyethylene-terephthalate nonwoven fabric using a blade coater. The composite cathode mixture was prepared by mixing CAM particles (85 wt %), LPSC (14.25 wt %), CF (0.25 wt %), and acrylate binder (0.5 wt %) in xylene with mortar. The composite cathode mixture was stretched into a 100- μm -thick sheet using a roll press. The SUS current collector, Ag–C sheet, SSE separating layer, composite cathode, and aluminum current collector were punched out. After stacking all the layers in a laminate bag and performing vacuum sealing, warm isostatic pressing was performed at 500 MPa and 85 $^{\circ}\text{C}$ for 30 min (EWIP-90-200, Ilshin Autoclave). Most of the assembly was performed in dry air at a dew point controlled to be below $-50\text{ }^{\circ}\text{C}$.

Electrochemical Analysis. EIS measurements of the electrochemical cells were performed using a VMP-3 potentiogalvanostat (BioLogic) in the frequency range of 1 MHz to 10 mHz with a fluctuating voltage of ± 20 mV. For the temperature-dependent experiments, the measurements were performed 45 min after the samples had stabilized at their corresponding temperatures in a temperature-controlled chamber (SU-222, ESPEC). The time-dependent variation in temperature was investigated using a temperature-controlled chamber (Figure S11). The resistances were normalized to the area specific resistance.

Battery tests were performed using a TOSCAT-3100 potentiogalvanostat (TOYO System) in a temperature-controlled chamber. The rate capability of the ASSB cell was evaluated by charging at a constant current (0.1 C-rate) followed by an upper-limit voltage until the current flow decreased to 0.05 C-rate. The cylindrical cells were discharged at a constant current (0.05–2 C-rate) in the voltage range 1.9–3.6 V versus the reduction potential of the anode (0.1 C-rate; 0.5 $\text{mA}\cdot\text{h}\cdot\text{cm}^{-2}$). For pouch-type high-energy ASSBs, the cell was discharged at a constant current (0.1–1 C-rate) in the voltage range 2.5–4.25 V (0.1 C-rate; 0.63 $\text{mA}\cdot\text{h}\cdot\text{cm}^{-2}$). A stacking pressure of 4 MPa was applied to the pouch cells during the battery tests.

Electrode Characterization. The morphologies of the composite cathodes were investigated by high-resolution SEM (SU8030 FE-SEM, Hitachi) at an applied voltage of 15 kV. For the cross-sectional imaging, the surfaces of the cathodes and the ASSB cells were polished using a cooling cross-section polisher (IB-19520CCP, JEOL). To prevent the contamination and reaction caused by air exposure, all samples were moved using a focused-ion-beam transfer vessel. The pore volume (ϵ_{pore}) of the composite cathode was determined as follows: $\epsilon_{\text{pore}} = (V_{\text{total}} - V_{\text{i}} - V_{\text{e}})/V_{\text{total}}$, where V_{total} , V_{i} , and V_{e} represent the volumes of the pellet, LPSC, and NCM, respectively.

Design of Equivalent Circuits and Fitting. The parameters of the circuit models were fit to the obtained impedance spectra using a spreadsheet (Excel Solver, Microsoft) to minimize the χ^2 value between the experimentally obtained and simulated profiles. The Kramers–Kronig test was performed to quantify the reliability of the experimentally obtained EIS data by verifying the stability, causality, and linearity of the impedance measurements (Figures S8–S10). The experimental data with relative errors within $\pm 1.0\%$ were utilized for fitting. The model circuit for full-cell-type ASSBs comprised the composite cathode (the TLM-based circuit in Figure 1d), SE separator ($R_{\text{SE}}^{\text{bulk}} + R_{\text{SE}}^{\text{gb}}/Q_{\text{SE}}^{\text{gb}}$), anode ($Q_{\text{anode}}/[R_{\text{anode}} + W_{\text{s}}]$), and cell inductance (L_{cell}). The electronic resistivity of the composite cathode (r_{e}) determined using the cell with the NCM-based composite cathode was assumed to be constant for all the cells. The total charge-transfer impedance ($Z_{\text{i/e}} [\Omega\cdot\text{cm}^2]$) was calculated as $z_{\text{i/e}}/t_{\text{cathode}}$, and the total impedance of the ion conduction ($Z_{\text{i}} [\Omega\cdot\text{cm}^2]$) was expressed as $z_{\text{i}}\cdot t_{\text{cathode}}$; $z_{\text{i/e}} [\Omega\cdot\text{cm}^3]$ represents the charge-transfer impedance per unit volume, and $t_{\text{cathode}} [\text{cm}]$ and $A_{\text{cathode}} [\text{cm}^2]$ represent the thickness and area of the cathode, namely, the cell dimensions.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsami.4c01322>.

Additional data, EIS results, SEM images, fitting parameters of EIS results, EIS simulation results, battery cell specification; Arrhenius plot of a symmetric cell (Figure S1); Nyquist plots of the composite cathodes and the corresponding fits obtained using a transmission-line-model (Figure S2); cross-sectional SEM images of the composite cathodes (Figures S3, S4, and S6); simulation of different Nyquist plots and schematic of shape changes of the cathodic impedance spectra of the composite cathode (Figure S5); energy density and capacity of the prototype cell as a function of stack number (Figure S7); relative errors of the real and imaginary parts of EIS data (Figures S8–S10); time-dependent temperature measurements performed in a temperature-controlled chamber (Figure S11); parameters used for fitting impedance spectra of the symmetric cells (Table S1); parameters used for fitting impedance spectra of a full-cell-type ASSB (Table S2); cell configuration of the prototype cell in single cell entry (Table S3); calculation of energy density of the prototype cell for different stack numbers (Table S4); parameters used for fitting impedance spectra of a pouch-type cell (Table S5) (PDF)

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<https://pubs.acs.org/10.1021/acsami.4c01322>

Author Contributions

The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript. W.C. contributed to conceptualization, investigation, visualization, writing – original draft, and writing – review and editing; J.H.K. contributed to methodology and investigation; Y.K. contributed to methodology; H.G. contributed to methodology; D.Y. contributed to methodology; G.Y. contributed to methodology; J.-S.K. contributed to supervision.

Notes

The authors declare no competing financial interest.

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